Water quality survey report: Sample Lake

AquaSentinel autonomous survey · last 1 h · generated 14 Oct 2026 08:53 IST · 225 readings

Simulated data. This report was generated from the built-in lake simulator for demonstration. It does not describe a real water body and must not be used for regulatory action.

1. Measured parameters

Parameter	Unit	Readings	Min	Mean	Max
pH	-	225	5.37	7.11	7.52
Dissolved oxygen	mg/L	225	4.86	6.59	7.15
Temperature	°C	225	28.1	28.9	31.6
Conductivity	µS/cm	225	386	632	1726
Nitrate	mg/L	225	2.8	3.8	5.3
Turbidity	NTU	225	8.3	14.0	31.4

Zone mean	pH	Dissolved oxygen	Temperature	Conductivity	Nitrate	Turbidity
N	7.44	6.97	28.7	411	4.2	11.1
NW	7.44	6.93	28.5	410	3.9	10.2
S	7.33	6.76	28.5	496	3.6	12.2
SE	7.45	6.78	28.3	432	3.5	12.0
SW	6.88	6.42	29.2	768	4.0	15.7

2. CPCB designated best use, from measured parameters

Lake-wide means meet the pH, DO and conductivity limits of **Class A**: Drinking water source without conventional treatment but after disinfection.

The worst readings in the window (pH 5.37, DO 4.86 mg/L, conductivity 1726 µS/cm) meet the limits of **no class**. A lake-wide mean can hide a local discharge, so both are shown.

Based on pH, DO and conductivity only. total coliform, BOD, free ammonia, SAR, boron are not measured, so the true class may be lower. Transcribed for this prototype. Check every value against the current CPCB publication before using a report for regulatory action.

3. Anomalies

#	Time (IST)	Zone	Parameters	Peak readings	Source estimate	Probable cause
1	14 Oct 2026 08:51	SW	Conductivity, pH, Temperature, Dissolved oxygen, Turbidity	Conductivity 1726, pH 5.37, Temperature 31.6, Dissolved oxygen 5.03, Turbidity 31.4	10.99746, 76.99729	industrial discharge (high)

Probable causes are written by a Gemini agent from the sensor and image evidence. They are leads for inspection and lab confirmation, not findings of fact.

4. Risk to aquatic life by zone (machine-learning model)

Zone	Status	Confidence	Reliable	Main factor
NW	Healthy	99%	yes	Daily temperature swing (24 h)
N	Healthy	99%	yes	Daily temperature swing (24 h)

Zone	Status	Confidence	Reliable	Main factor
NE	Healthy	99%	yes	Daily pH swing (24 h) - algal bloom indicator
SW	Healthy	56%	yes	Daily pH swing (24 h) - algal bloom indicator
S	Healthy	94%	yes	pH maximum, last 1h
SE	Healthy	99%	yes	Time of day

LightGBM fish-health classifier trained on simulated pond data for tilapia and Indian major carps. It indicates risk to aquatic life from temperature, TDS, turbidity and pH patterns; it does not measure dissolved oxygen or ammonia and has not been validated on lake field data.

5. Narrative

Written by Gemini (gemini-3.5-flash-lite) from the data above. Check it against the tables before relying on it.

Summary

This survey report covers the last 1 hours of monitoring data collected by the autonomous boat in Sample Lake. Data originate from the built-in simulator. Lake-wide water quality is generally stable, but a significant pollution event was detected in the south-west zone near the South-west drain.

Findings by zone

- **North-west (NW):** pH mean 7.44 (range 7.40–7.48), DO mean 6.93 mg/L (range 6.80–7.11 mg/L), temperature mean 28.53 °C, EC mean 410.32 µS/cm, nitrate mean 3.87 mg/L, turbidity mean 10.19 NTU across 9 readings.
- **North (N):** pH mean 7.44 (range 7.43–7.46), DO mean 6.97 mg/L, temperature mean 28.66 °C, EC mean 411.50 µS/cm, nitrate mean 4.17 mg/L, turbidity mean 11.05 NTU across 3 readings (limited sample size).
- **North-east (NE):** pH mean 7.45, DO mean 6.78 mg/L, temperature mean 28.28 °C, EC mean 432.46 µS/cm, nitrate mean 3.50 mg/L, turbidity mean 11.96 NTU across 31 readings.
- **South (S):** pH mean 7.33 (min 6.61, max 7.51), DO mean 6.76 mg/L, temperature mean 28.49 °C, EC mean 496.13 µS/cm (max 937.53 µS/cm), nitrate mean 3.62 mg/L, turbidity mean 12.23 NTU across 59 readings (15 flagged anomalies).
- **South-west (SW):** pH mean 6.88 (min 5.37, max 7.52), DO mean 6.42 mg/L (min 4.86 mg/L), temperature mean 29.21 °C (max 31.56 °C), EC mean 770.54 µS/cm (max 1726.20 µS/cm), nitrate mean 4.01 mg/L (max 5.29 mg/L), turbidity mean 15.75 NTU (max 31.36 NTU) across 122 readings (73 flagged anomalies).
- **South-east (SE):** pH mean 7.45, DO mean 6.78 mg/L, temperature mean 28.28 °C, EC mean 432.46 µS/cm, nitrate mean 3.50 mg/L, turbidity mean 11.96 NTU across 31 readings.

Anomalies and probable sources

One major hotspot anomaly was detected and traced in the South-west zone, located 4 m from the South-west drain. Peak readings at this location showed electrical conductivity surging to 1726.20 µS/cm (baseline 416.28 µS/cm), pH dropping to 5.37 (baseline 7.45), temperature rising to 31.56 °C, turbidity increasing to 31.36 NTU, and dissolved oxygen dropping to 5.03 mg/L. Visual inspection confirmed distinct rainbow iridescent patches (oil sheen) on the water surface. The probable cause is industrial discharge.

CPCB assessment

Lake-wide average parameters satisfy CPCB Designated Best Use Class A (Drinking water source without conventional treatment but after disinfection). However, at the peak anomaly location in the South-west zone, water quality falls below Class E due to acidic pH (5.37).

Risk to aquatic life

The fish-health machine-learning model rates all zones as "Healthy" with confidence ranging from 0.56 to 0.99. However, the model does not measure dissolved oxygen or toxic chemical parameters, and operator advice notes potential risks from low pH and high TDS in the South-west zone.

Recommended actions

- Dispatch an enforcement team to inspect the South-west drain and surrounding industrial units.

- Collect grab samples near the source trace location for heavy metal, oil and grease, COD, and TDS analyses.
- Deploy containment booms immediately to capture and absorb the oil sheen.

Limitations of this survey

The figures are simulated rather than from a field survey. The boat does not measure total coliform, biological oxygen demand (BOD), free ammonia, SAR, or boron. The fish-health model has not been validated on lake field data and does not evaluate dissolved oxygen or nitrate levels. Data from the North zone are based on a very limited number of readings (3).